Shipping System — UML Class Diagram

10 Design Patterns · com.shipping

Legend
- solid ■ Inheritance (extends)
- dashed ■ Realization (implements)
- → Dependency / uses
- ■→ Aggregation (holds ref)

1 · Facade · Builder · Singleton · Core Model

«facade» ShippingService
- -vehicleService:VehicleService
- -dispatcher:VehicleDispatcher
- -context:ShippingContext
- +calculatePrice(o):double
- +getAvailableVehicles(o):List
- +getPackaging(o):Packaging
- +dispatchOrder(o):void
- +registerCourier(c):void

«builder» «model» ShippingOrder
- -country:String
- -itemType:String
- -weight:double
- -distance:double
- -deliveryDate:String
- -vehicleType:String
- +getCountry():String
- +getWeight():double
- +getDistance():double
- +getVehicleType():String
- «inner» Builder

Vehicle
- -type:String
- -allowedCountries:List<String>
- -maxCapacity:double
- +getType():String
- +getAllowedCountries():List
- +getMaxCapacity():double

«singleton» PricingConfig
- -instance:PricingConfig {static}
- -pricePerKg:double=2.5
- -pricePerKm:double=0.5
- +getInstance():PricingConfig {static}
- +getPricePerKg():double
- +getPricePerKm():double

2 · Chain of Responsibility — Pricing Pipeline

«abstract» PricingHandler
- #next:PricingHandler
- +setNext(h):PricingHandler
- +handle(order,price):double*

WeightPricingHandler
- +handle(order,price):double
  price += weight × perKg

DistancePricingHandler
- +handle(order,price):double
  price += distance × perKm

DatePricingHandler
- +handle(order,price):double
  EXPRESS → price × 1.5

ItemTypePricingHandler
- +handle(order,price):double
  FRAGILE+20 | OVERSIZED+50

next →

3 · Factory Method — Packaging

creates

«interface» Packaging
- +getType():String
- +getDescription():String

«abstract» PackagingFactory
- +createPackaging():Packaging*
- +preparePackaging():String

StandardPackagingFactory
- +createPackaging():Packaging
  returns: Standard cardboard box

FragilePackagingFactory
- +createPackaging():Packaging
  returns: Bubble-wrap box

LiquidPackagingFactory
- +createPackaging():Packaging
  returns: Liquid-safe sealed box

4 · Decorator — Courier

«interface» Courier
- +getName():String
- +deliver(order):String

BasicCourier
- -name:String
- +getName():String
- +deliver(order):String

«abstract» CourierDecorator
- #wrapped:Courier
- +getName():String
- +deliver(order):String

ReviewableCourier
- -reviews:List<String>
- -totalRating:double
- -ratingCount:int
- +addReview(r,rating):void
- +getAverageRating():double
- +deliver(order):String

5 · Observer — Proxy — Strategy

«interface» VehicleSubject
- +addObserver(o)…
- +removeObserve…
- +notifyObserver…

notifies

VehicleDispatcher
- -observers:List…
- +dispatch(type…
- +notifyObserver…

«interface» CourierObserver
- +onVehicleDispa…

AvailableCourier
- -name:String
- +onVehicleDispa…

«interface» VehicleService
- +getAvailableVehicles(…

RealVehicleService
- -allVehicles:List<Veh…
- +getAvailableVehicles(…
  applies strategies

delegates

VehicleServiceProxy
- -real:RealVehicleServ…
- -cache:Map<String,List>
- +getAvailableVehicles(…
  cache → real if miss

«interface» SelectionStrategy
- +select(vehicles,orde…

ByCountryStrategy
- +select(vehicles,orde…
  filter: allowedCountr…

ByWeightStrategy
- +select(vehicles,orde…
  filter: maxCapacity …

6 · State — Region Shipping Rules

«interface» RegionState
- +getAllowedVehicles():List
- +getMaxWeight():double
- +getRequiredDocumentation():String

state

ShippingContext
- -state:RegionState
- +setState(s):void
- +getAllowedVehicles():List
- +getMaxWeight():double

EgyptState
- Vehicles:TRUCK,VAN,PLANE,DRONE
- MaxWeight:10000 kg
- Doc:Customs form EG-01

UAEState
- Vehicles:PLANE,SHIP,DRONE
- MaxWeight:10000 kg
- Doc:GCC clearance

SaudiArabiaState
- Vehicles:PLANE,SHIP,DRONE
- MaxWeight:10000 kg
- Doc:GCC clearance

EuropeState
- Vehicles:PLANE,SHIP
- MaxWeight:10000 kg
- Doc:EU customs declaration

Design Patterns Summary

Singleton	Builder	Facade	Chain of Responsibility	Factory Method	Decorator	Observer	Proxy	Strategy	State
PricingConfig	ShippingOrder.Builder	ShippingService	PricingHandler chain	PackagingFactory	CourierDecorator	VehicleDispatcher	VehicleServiceProxy	SelectionStrategy	RegionState